Md Abir Hossain

+44 7507034017 | abirfarabi915@gmail.com | linkedin.com/in/abir-farabi | Github | Birmingham, UK

Summary

MSc graduate in Artificial Intelligence and Machine Learning from the University of Birmingham. Hands-on experience building machine learning & AI models across multiple domains including generative AI, RAG, computer vision, data science, statistical analysis, time series forecasting and deep learning. Currently seeking AI Engineer roles.

Experience

Data Science Intern Mar. 2022 – Sep. 2022

Aziz & Brothers *Dhaka, Bangladesh*

- Analysed **50,000**+ customer records using SQL and Python to identify the top 3 best-selling products, enabling the business to negotiate cheaper procurement and generating **100,000** BDT in cost savings
- Cleaned and processed 50,000+ customer records, applying LSTM-based time series forecasting across customer segments and product categories to predict future sales, enabling targeted discount strategies that drove an estimated **15%** increase in sales
- Visualised monthly sales trends and product performance using Python and Matplotlib, identifying a 2-3% month-on-month decline over 7 months and presenting findings to management, contributing to strategic changes that reversed the trend to 4% growth by end of internship

Education

University of Birmingham Birmingham, UK

Master of Science in Artificial Intelligence and Machine Learning *Sep. 2023 – Sep. 2024*

- Modules: Intelligent Data Analysis, Machine Learning, Mathematical Foundations of AI & ML

United International University Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering *Jan. 2017 – Apr. 2022*

- Modules: Database Management System (82%), Pattern Recognition (90%), Artificial Intelligence (88%)

Projects

Sequential LLM Intelligence Pipeline | *Python, LangChain (LCEL), OpenAI, Pinecone, Streamlit*

- Engineered a multi-stage LLM reasoning pipeline using LangChain Expression Language (LCEL) and OpenAI APIs, implementing RAG with Pinecone vector database for contextual retrieval and entity extraction
- Implemented persistent session-state memory using ConversationBufferMemory and evaluated pipeline outputs rigorously, iterating on prompt design to improve accuracy and reliability across non-linear interactions

DataChat (In Development) | *Python, LangGraph, Claude API, FastAPI, Streamlit, Langfuse, Docker, AWS*

- Building an agentic data analyst application enabling users to query CSV datasets using natural language, with autonomous statistical analysis and visualisation generation
- Implementing multi-agent architecture using LangGraph with LLM observability and tracing via Langfuse

SlotFinder | *Python, LSTM, Pandas, Matplotlib*

- Co-authored and published research at **IEEE ICCIT 2022**, predicting vacant parking space departure and arrival times across 408 buildings in 7 Dhaka locations, reducing average parking search time from **20** minutes to **6-9** minutes
- Conducted 6-experiment ablation study achieving RMSE 0.29, MAE 0.24 with Adam (lr=0.01, batch=16), outperforming RNN (0.45), ARIMA (0.40) and LSTM (0.35)

Technical Skills

Languages: Python and SQL.
AI & ML: RAG, Agentic AI, Vector Database(Pinecone), Linear Regression, Logistic Regression, LLM(OpenAI), LSTM, CNN, RNN, XGBoost, Random Forest, A/B Testing.
Libraries & Frameworks: Scikit-learn, Pandas, NumPy, PyTorch, Hugging Face, LangChain, LangGraph.
MLOps & Tools: Databricks, Generative AI, Docker, MLflow, CI/CD pipeline, FastAPI, Streamlit, Git, Jupyter Notebook, AWS.